What is Apache Maven?

Apache Maven is a build automation and project management utility commonly used in Java development. It simplifies the build process, dependency management, and documentation of projects following the "convention over configuration" principle to reduce setup time.

Practical No. 8

Installation and Project Set Up

1. Install JDK on Windows 11
    a. Set system variables
    b. Press Windows Key
    c. Search → Environment Variables
    d. Click → *Edit the system environment variables*
    e. Click → Environment Variables
    f. Under *System Variables* → Click New
    g. Add: Variable Name: JAVA_HOME and Variable Value: C:\Program Files\Java\jdk-17
    h. (Adjust path as per your installation)
    i. Add Java to PATH
    j. Select Path
    k. Click Edit
    l. Click New
    m. Add: %JAVA_HOME%\bin
    n. Click OK
    o. Verify Java
    p. Open Command Prompt (cmd):
        i. java -version
        ii. javac -version
    q. You should see version details.
2. Install Maven on Windows 11
3. Download Maven from: https://maven.apache.org/download.cgi
   Choose: Binary zip archive
4. Extract Maven
5. Inside it should contain: bin, conf, lib
6. Set system variables for maven as well
    a. Go to Environment Variables again
    b. Click New
    c. Add: Variable Name: MAVEN_HOME and Variable Value: C:\apache-maven
    d. Add Maven to PATH
    e. Edit Path, Add: %MAVEN_HOME%\bin
    f. Click OK.
    g. Verify Maven
    h. Open new Command Prompt:
    i. mvn -version

    j. You should see: Maven version, Java version, OS: Windows 11
7. Create a folder for your project and open cmd from this folder, Enter following command to start creating structure of your project
    a. mvn archetype:generate
    b. When prompted:
    c. Choose: maven-archetype-quickstart or default
    d. Choose: version or set it to latest
    e. Enter:
        i. GroupId: com.college
        ii. ArtifactId: MavenDemo
        iii. Version: 1.0-SNAPSHOT
        iv. Package: com.college
8. Project created successfully.
9. System will ask for confirmation press: Y
10. Open your folder inside this MavenDemo Project is created
11. This folder contains src and pom.xml file
12. Src folder contains main and test folders
13. In \main\java\org\myapp\com\ App.java file is created you can open and check the same
14. Switch back to cmd and run Maven Commands
    a. mvn compile
    b. mvn test
    c. mvn package
15. Check: target folder
16. You will see: MavenDemo-1.0-SNAPSHOT.jar
17. Now from your Maven project folder open cmd and enter following command to execute the project

```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Chetna\myapp\MavenDemoApp>java -cp target/MavenDemoApp-1.0-SNAPSHOT.jar org.myapp.com.App
Hello World!

C:\Users\Chetna\myapp\MavenDemoApp>
```

Practical No. 9

Creating web application with maven

Open cmd from your folder where you want to save your web app

Then enter following command

mvn archetype:generate

Note: Copy the list of archetypes and paste in notepad then search for webapp-j2ee and note the id

Select achetype

```
2654: remote -> org.codehaus.mojo.archetypes:webapp-j2ee13 (J2EE 1.3 web application archetype)
2655: remote -> org.codehaus.mojo.archetypes:webapp-j2ee14 (J2EE 1.4 web application archetype)
2656: remote -> org.codehaus.mojo.archetypes:webapp-javaee6 (Archetype for a web application using Java EE 6.)
2657: remote -> org.codehaus.mojo.archetypes:webapp-javaee7 (Archetype for a web application using Java EE 7.)
```

Select the version and enter details as below:

```
Define value for property 'groupId': org.mywebapp.com
Define value for property 'artifactId': MyWebApp
Define value for property 'version' 1.0-SNAPSHOT:
Define value for property 'package' org.mywebapp.com:
Confirm properties configuration:
groupId: org.mywebapp.com
artifactId: MyWebApp
version: 1.0-SNAPSHOT
package: org.mywebapp.com
 Y: Y
```

In your chosen folder you will find a folder MyWebApp

Enter following command to change the directory

cd MyWebApp

Compile and package the default web app with following commands

mvn compile

mvn package

Now in MyWebApp folder you will find a target folder and in target folder MyWebApp-1.0-SNAPSHOT.war file

Now come back to MyWebApp and open src folder

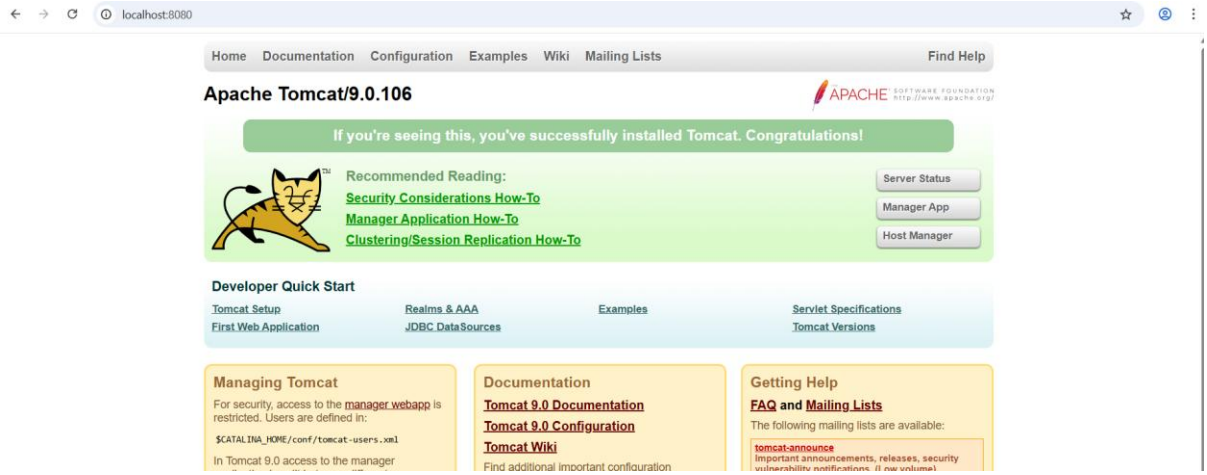
Go to \main\webapp you will see default index.jsp file

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
   "http://www.w3.org/TR/html4/loose.dtd">

<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>
        <h1>Hello World!</h1>
    </body>
</html>
```

Now we will deploy our war file with tomcat server

Open \apache-tomcat-9.0.106\bin\ and open startup.bat file then open browser and type localhost:8080



Click on Manage App

It will ask you to enter username and password, to check the same go to \apache-tomcat-9.0.106\conf\ and open tomcat-users.xml file with notepad check the username and set the password for the same also add user role and remove the comment tag

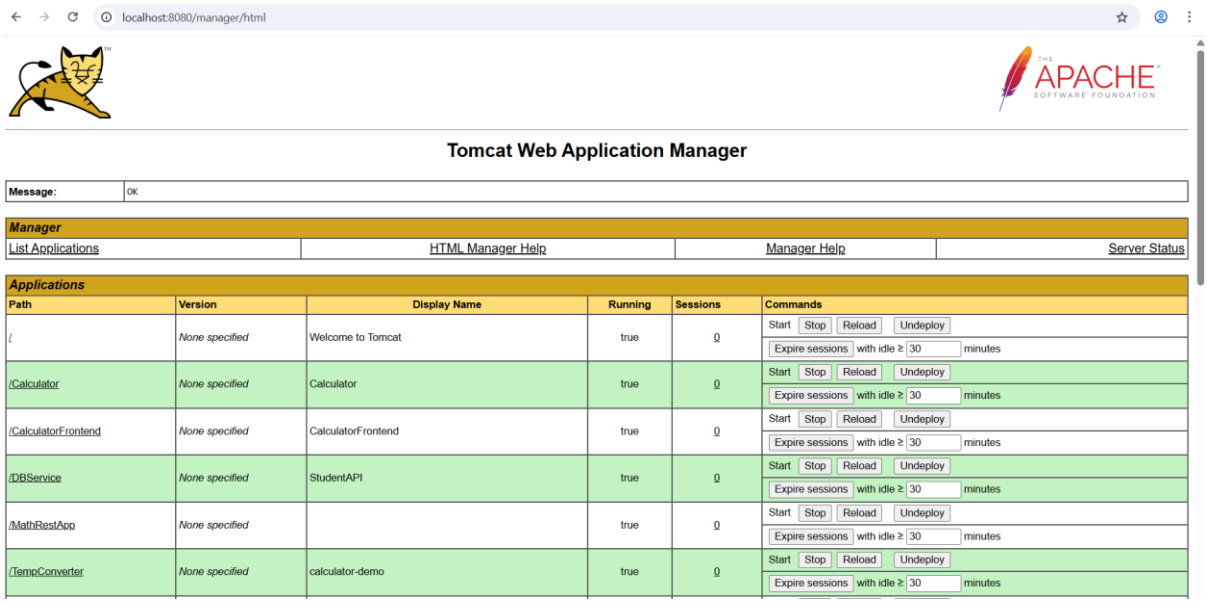
```
<!--
  <user username="admin" password="<must-be-changed>" roles="manager-gui"/>
  <user username="robot" password="<must-be-changed>" roles="manager-script"/>
-->
```

Replace the above code with following:

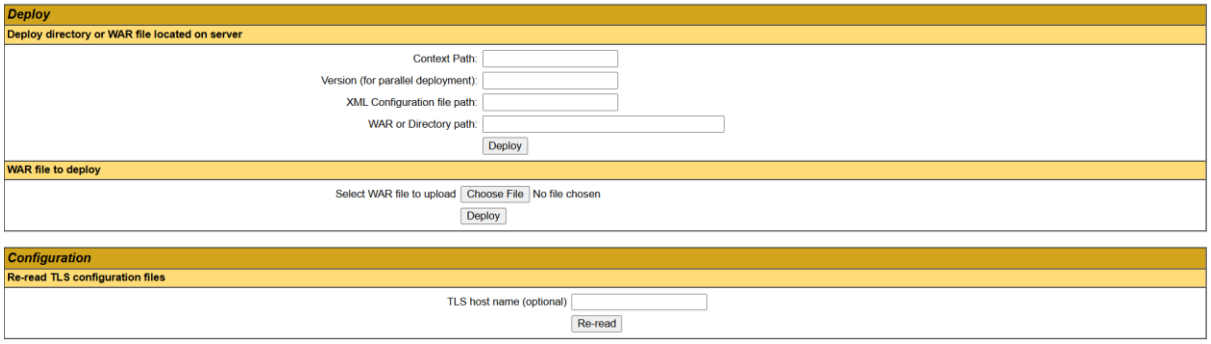
```
    <role rolename="manager-gui"/>
    <user username="admin" password="12345" roles="manager-gui"/>
```

Now stop and then restart tomcat server go to browser and refresh localhost:8080 and click on Manage App

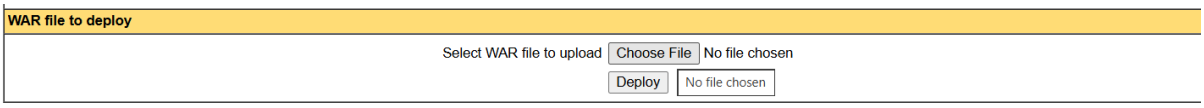
Enter username and password it will open



Scroll Down

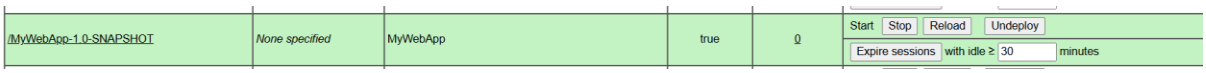


Click on choose file button:

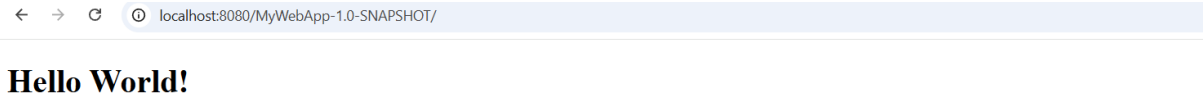


Go to \myapp\MyWebApp\target\ and select .war file and click on deploy button

After this you can find your .war file in the above list



Just click on it and you will see the output:



Practical No. 10

Introduction to maven plugins

A. Using maven compiler plugin

Open your pom.xml and notice the version under configuration

```
▼<build>
  ▼<plugins>
    ▼<plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>2.0.2</version>
      ▼<configuration>
          <source>1.4</source>
          <target>1.4</target>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>
```

It is 1.4 since we have created j2ee 4 project

Now we will add generic code to this project which can't be compiled with 1.4

Then we will add compiler with configuration 1.5

To add a code go to \MyWebApp\src\main\java\org\mywebapp\com and create one java file there as MyClass.java

```
J MyClass.java 2 ●
C: > Users > Chetna > myapp > MyWebApp > src > main > java > org > mywebapp > com > J MyClass.java > MyClass >
1  package org.mywebapp.com;
2
3  import java.util.*;
4
5  public class MyClass
6  {
       Run | Debug
7      public static void main(String[] args)
8      {
9          List<String> stringList = new ArrayList<String>();
10     }
11 }
```

Now let's try to compile this

```
C:\Users\Chetna\myapp\MyWebApp>mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< org.mywebapp.com:MyWebApp >----------------------
[INFO] Building MyWebApp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ MyWebApp ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Chetna\myapp\MyWebApp\src\main\resources
[INFO]
[INFO] --- compiler:2.0.2:compile (default-compile) @ MyWebApp ---
[INFO] Compiling 1 source file to C:\Users\Chetna\myapp\MyWebApp\target\classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD FAILURE
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.574 s
[INFO] Finished at: 2026-02-21T19:16:42+05:30
[INFO] ------------------------------------------------------------------------
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:2.0.2:compile (default-compile) on project
 MyWebApp: Compilation failure: Compilation failure:
[ERROR] C:\Users\Chetna\myapp\MyWebApp\src\main\java\org\mywebapp\com\MyClass.java:[9,12] error: generics are not suppor
ted in -source 1.4
[ERROR]
[ERROR] could not parse error message:   (use -source 5 or higher to enable generics)
[ERROR] C:\Users\Chetna\myapp\MyWebApp\src\main\java\org\mywebapp\com\MyClass.java:9: error: '(' or '[' expected
[ERROR]         List<String> stringList = new ArrayList<String>;
```

We have got error here which clearly indicate we can't compile this java file with version 4, we need either version 5 or higher

Now to change the version open pom.xml file and edit it as below

```
  </dependencies>
  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>2.0.2</version>
        <configuration>
          <source>1.5</source>
          <target>1.5</target>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>
```

Now again try to compile the project

```
C:\Users\Chetna\myapp\MyWebApp>mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< org.mywebapp.com:MyWebApp >----------------------
[INFO] Building MyWebApp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ MyWebApp ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Chetna\myapp\MyWebApp\src\main\resources
[INFO]
[INFO] --- compiler:2.0.2:compile (default-compile) @ MyWebApp ---
[INFO] Compiling 1 source file to C:\Users\Chetna\myapp\MyWebApp\target\classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.065 s
[INFO] Finished at: 2026-02-21T19:22:57+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\Chetna\myapp\MyWebApp>
```

Since maven 3 uses compiler 5 by default if we remove the build section from pom.xml and try to clean and compile again it will get build successfully

B. Using server plugin (tomcat or jetty)

Now many times we want to deploy our web application on server or any container.

Maven provides some server and container plugins to deploy our web application.

We have plugins for tomcat and jetty, these are light weight server containers

For jetty, declare it in pom.xml

```
▼<build>
  ▼<plugins>
    ▼<plugin>
       <groupId>org.eclipse.jetty</groupId>
       <artifactId>jetty-maven-plugin</artifactId>
       <version>9.4.53.v20231009</version>
     </plugin>
   </plugins>
 </build>
```

For Tomcat

```
▼<build>
  ▼<plugins>
    ▼<plugin>
       <groupId>org.apache.tomcat.maven</groupId>
       <artifactId>tomcat7-maven-plugin</artifactId>
       <version>2.2</version>
       <!--  Use the appropriate version  -->
     </plugin>
   </plugins>
 </build>
```

Enter to following command to start with jetty server

mvn jetty:run

To start with tomcat server

mvn tomcat7:run

Then open the browser and type localhost:8080 (for jetty) or localhost:8080/yourWebApp (for tomcat)

Hello World!

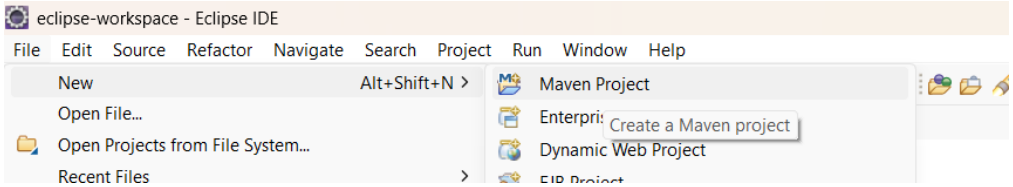
Practical No. 11

Creating maven project in eclipse

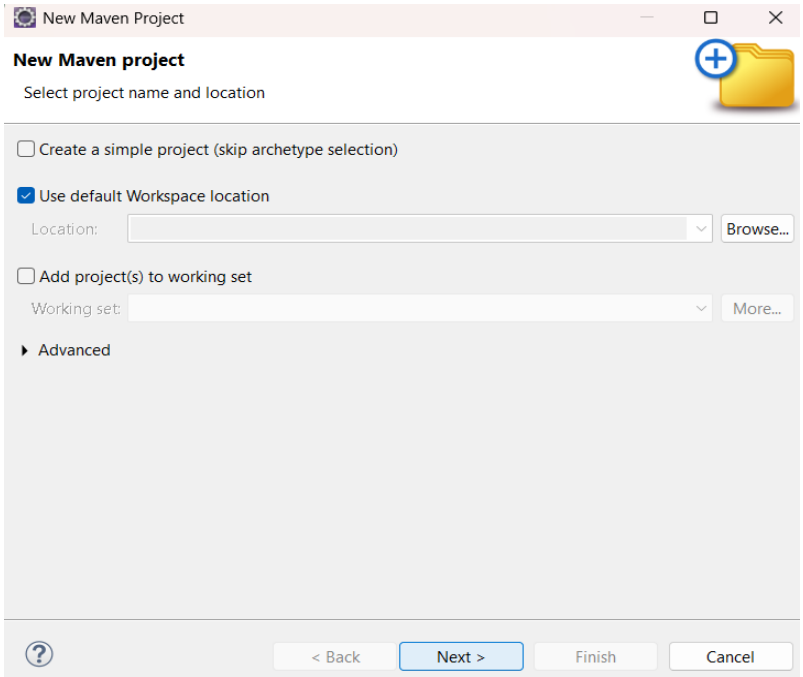
A. Creating simple Hello World! Web application with maven

Open eclipse

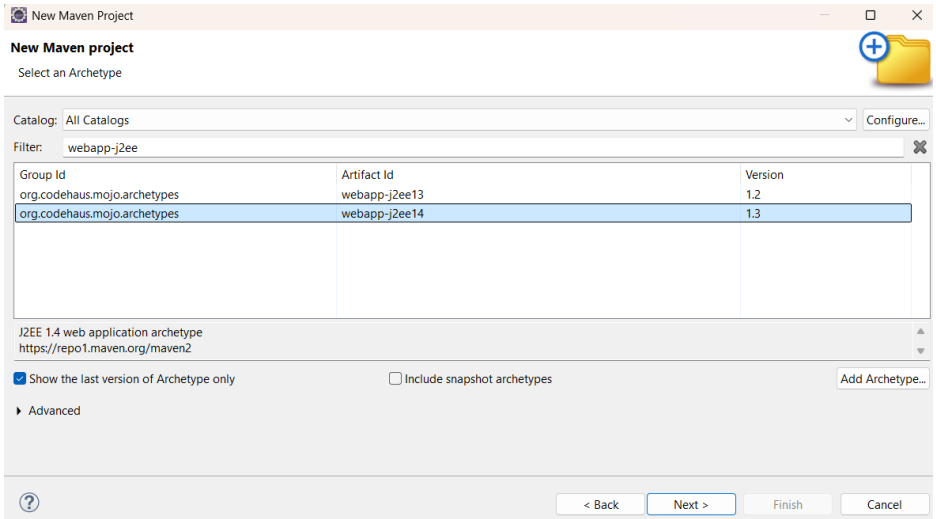
Click on file → new → Maven Project



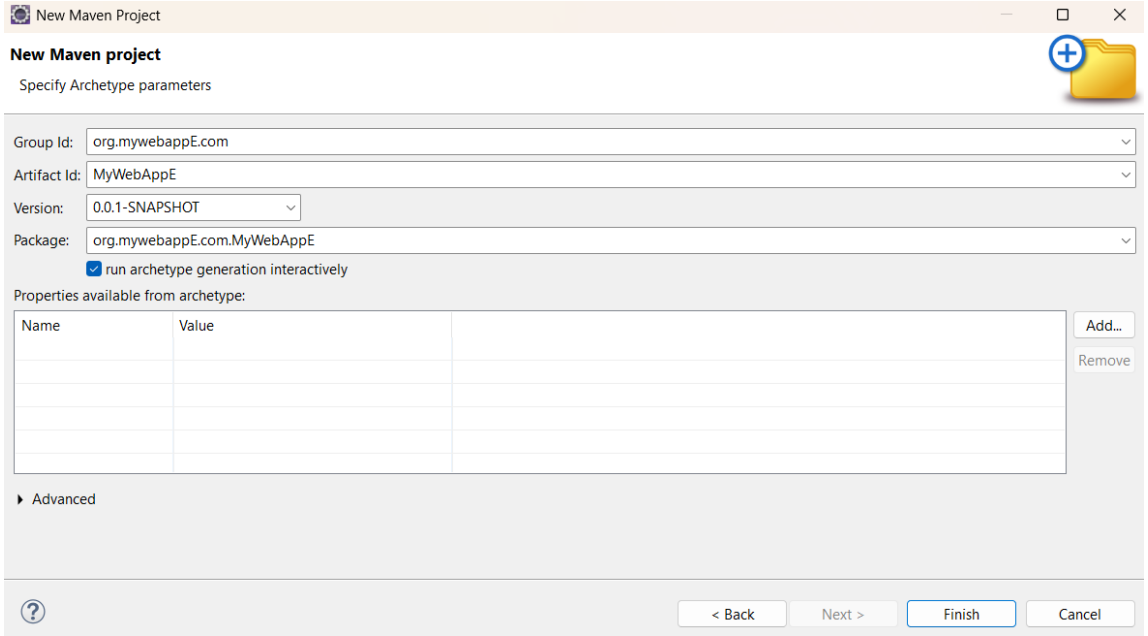
Check the checkbox if you want to create a default project and don't want to select archetype otherwise keep it unchecked and click on next



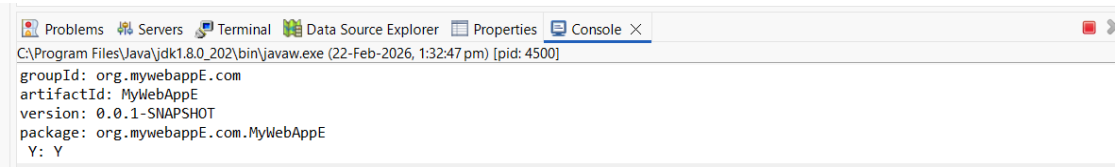
Enter webapp-j2ee in filter select the archetype and click on next



Enter the group id, artifact id, snapshot and package then click on finish



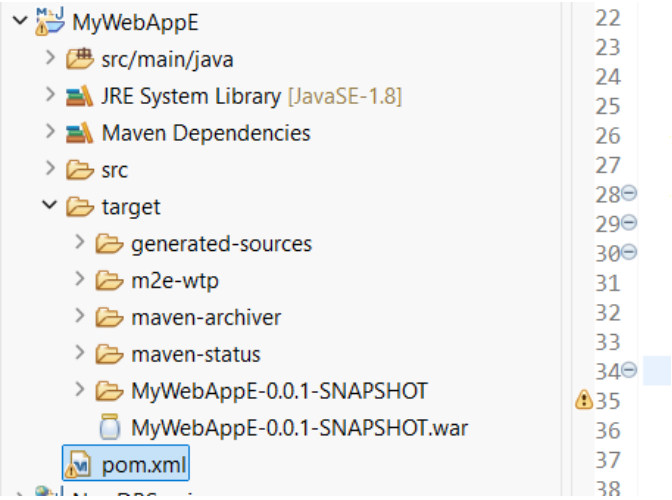
In the console type Y for confirmation



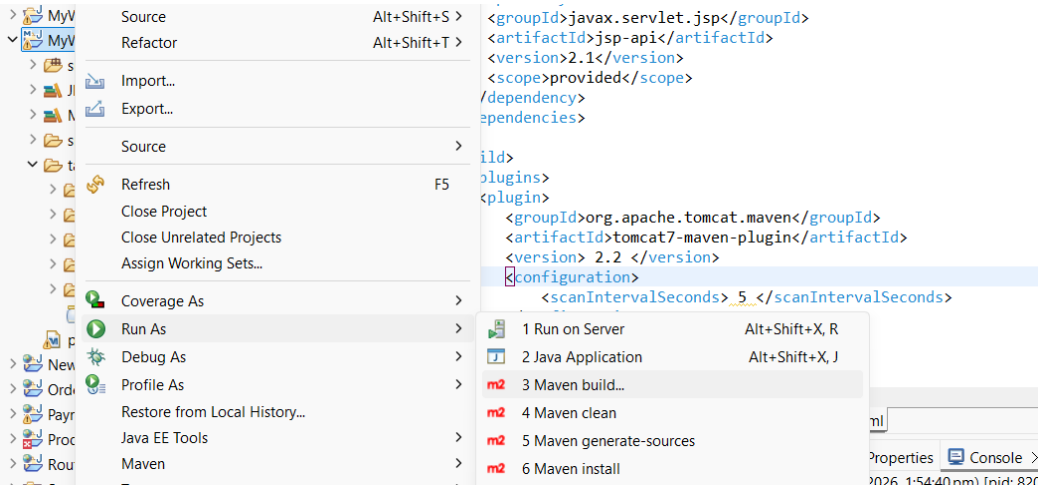
Maven project is created

Right click on your project click on run as → Maven Install

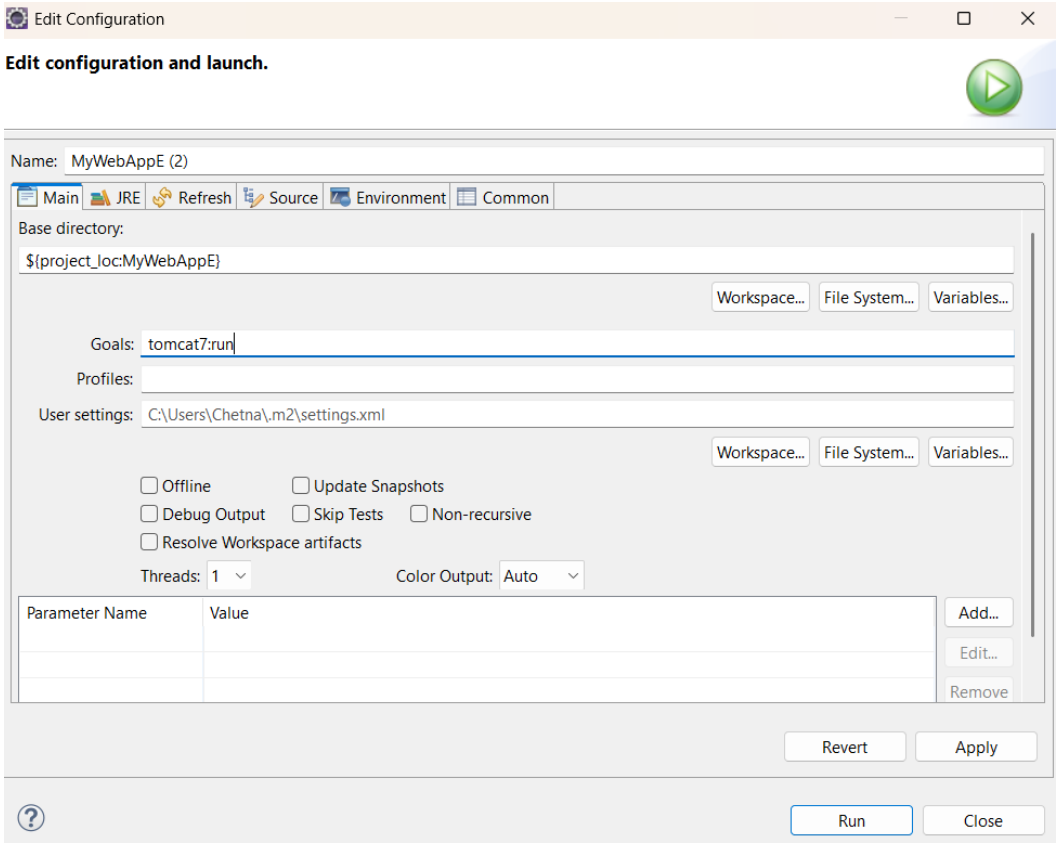
After it is done check the target folder under your project it should contain .war file



Again right click on project and select run as Maven Build…



Enter goal



Click on run and once server started

Open browser and enter localhost:8080/MyWebAppE

B. Creating simple Calculator Web application with maven

Follow the above steps to create a maven project and then edit pom.xml file to add server plugin (jetty or tomcat)

Edit index.jsp file as follow:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
   "http://www.w3.org/TR/html4/loose.dtd">

<html>
<head>
<title>Calculator App</title>
</head>
<body>
<form action="CalculatorServlet" method="get" >
Enter First Number: <input type="text" name="txtN1" ><br>
Enter Second Number: <input type="text" name="txtN2" ><br>
Select an Operation
<input type="radio" name="opr" value="+">ADDTION
<input type="radio" name="opr" value="-">SUBSTRACTION
<input type="radio" name="opr" value="*">MULTIPLY
<input type="radio" name="opr" value="/">DIVIDE <br>
<input type="reset">
<input type="submit" value="Calculate" >
</form>
</body>
</html>
```

Add a servlet file to the project as CalculatorServlet.java

```
package org.mywebappE.com.MyWebAppE;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
public class CalculatorServlet extends HttpServlet
{
public void doGet(HttpServletRequest request, HttpServletResponse response)throws ServletException, IOException
    {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        out.println("<html><head><title>Calculator Servlet</title></head><body>");
        double n1 = Double.parseDouble(request.getParameter("txtN1"));
        double n2 = Double.parseDouble(request.getParameter("txtN2"));
        double result =0;
        String opr=request.getParameter("opr");
        if(opr.equals("+"))
            result=n1+n2;
        if(opr.equals("-"))
            result=n1-n2;
        if(opr.equals("*"))
            result=n1*n2;
        if(opr.equals("/"))
            result=n1/n2;
        out.println("<h1> Result = "+result);
        out.println("</body></html>");
    }
}
```

Save the project, build and execute in browser

localhost:8080/MyWebAppE/

Enter First Number: 1

Enter Second Number: 2

Select an Operation ADDTION SUBSTRACTION MULTIPLY DIVIDE

Reset Calculate

localhost:8080/MyWebAppE/CalculatorServlet?txtN1=1&txtN2=2&opr=%2B

Result = 3.0

Reference:

https://youtube.com/playlist?list=PL92E89440B7BFD0F6&si=5cgMGfsMjBucZ-ix